

Unit 5: Continuity — Full Solutions

Warm-up

1. $f(x) = x^2 + 3$ at $x = 2$.

Check 1 — is $f(2)$ defined? $f(2) = 4 + 3 = 7$. Yes, defined.

Check 2 — does the limit exist? Polynomials are smooth everywhere, so $\lim_{x \rightarrow 2} f(x) = 2^2 + 3 = 7$. Yes, exists.

Check 3 — does the limit equal $f(2)$? $7 = 7$. Yes.

Answer: f is continuous at $x = 2$. (This will always be true for polynomials — they're continuous everywhere.)

2. $f(x) = \frac{x+1}{x-2}$ at $x = 0$ and $x = 2$.

At $x = 0$: $f(0) = \frac{1}{-2} = -\frac{1}{2}$, defined. Since this is a rational function and the denominator isn't zero at $x = 0$, the limit equals $f(0)$ automatically.

Answer: continuous at $x = 0$.

At $x = 2$: the denominator becomes $2 - 2 = 0$, so $f(2)$ is undefined. Condition 1 already fails.

Answer: discontinuous at $x = 2$. Since the numerator at $x = 2$ is $2 + 1 = 3 \neq 0$, the function blows up toward $\pm\infty$ near $x = 2$ — this is an infinite discontinuity (a vertical asymptote).

3. $f(x) = \begin{cases} x+2, & x < 1 \\ 4, & x \geq 1 \end{cases}$ at $x = 1$.

$f(1) = 4$ (defined, using the second piece).

Left-hand limit: $1 + 2 = 3$. Right-hand limit: 4.

Since $3 \neq 4$, the limit does not exist.

Answer: discontinuous at $x = 1$ (condition 2 fails — this is a jump discontinuity).

4. $f(x) = \begin{cases} x^2, & x \leq 2 \\ 2x, & x > 2 \end{cases}$ at $x = 2$.

$f(2) = 2^2 = 4$ (defined, using the first piece since $x \leq 2$ includes 2).

Left-hand limit: $2^2 = 4$. Right-hand limit: $2(2) = 4$. Both match, so the limit exists and equals 4.

Compare to $f(2) = 4$: they match.

Answer: f is continuous at $x = 2$. All three conditions check out.

5. $f(x) = \begin{cases} 7, & x = 0 \\ x^2 + 3, & x \neq 0 \end{cases}$ at $x = 0$.

$f(0) = 7$, defined.

The limit uses the “ $x \neq 0$ ” piece (valid on both sides near 0): $\lim_{x \rightarrow 0} (x^2 + 3) = 0 + 3 = 3$. So the limit exists and equals 3.

Compare to $f(0) = 7$: they do not match ($3 \neq 7$).

Answer: discontinuous at $x = 0$. This is a removable discontinuity — the limit exists just fine (it’s heading toward 3), but the function has been deliberately defined to the “wrong” value (7) right at that point. Redefining $f(0) = 3$ would patch the hole completely.

6. $f(x) = \frac{1}{x-3}$ at $x = 3$.

$f(3)$ would require dividing by 0, so $f(3)$ is undefined.

Answer: discontinuous at $x = 3$. Since the numerator (1) never becomes zero, the function shoots off toward $\pm\infty$ as $x \rightarrow 3$ — this is an infinite discontinuity (a vertical asymptote at $x = 3$).

Standard

7. $f(x) = \begin{cases} x^2 - 2x + 3, & x < 3 \\ 2ax, & x \geq 3 \end{cases}$, find a for continuity at $x = 3$.

Left-hand limit (use $x^2 - 2x + 3$): $3^2 - 2(3) + 3 = 9 - 6 + 3 = 6$.

Right-hand value/limit (use $2ax$): $2a(3) = 6a$.

Set them equal for continuity:

$$6a = 6$$

$$a = 1$$

Answer: $a = 1$.

8. $f(x) = \begin{cases} kx + 1, & x < 2 \\ x^2 - 1, & x \geq 2 \end{cases}$, find k for continuity at $x = 2$.

Left-hand limit: $k(2) + 1 = 2k + 1$.

Right-hand value: $2^2 - 1 = 3$.

Set equal:

$$2k + 1 = 3$$

$$2k = 2$$

$$k = 1$$

Answer: $k = 1$.

9. $f(x) = \begin{cases} 3x + b, & x \leq 1 \\ x^2 + 2, & x > 1 \end{cases}$, find b for continuity at $x = 1$.

Left-hand value (using the piece that includes $x = 1$): $3(1) + b = 3 + b$.

Right-hand limit: $1^2 + 2 = 3$.

Set equal:

$$3 + b = 3$$

$$b = 0$$

Answer: $b = 0$.

10. $f(x) = \begin{cases} x^2 + 4, & x < -1 \\ mx - 3, & x \geq -1 \end{cases}$, find m for continuity at $x = -1$.

Left-hand limit: $(-1)^2 + 4 = 1 + 4 = 5$.

Right-hand value: $m(-1) - 3 = -m - 3$.

Set equal:

$$-m - 3 = 5$$

$$-m = 8$$

$$m = -8$$

Answer: $m = -8$.

11. $f(x) = \begin{cases} ax^2, & x \leq 2 \\ 4x - 4, & x > 2 \end{cases}$, find a for continuity at $x = 2$.

Left-hand value: $a(2)^2 = 4a$.

Right-hand limit: $4(2) - 4 = 8 - 4 = 4$.

Set equal:

$$4a = 4$$

$$a = 1$$

Answer: $a = 1$.

12. $f(x) = \begin{cases} x^3, & x < 1 \\ c - x, & x \geq 1 \end{cases}$, find c for continuity at $x = 1$.

Left-hand limit: $1^3 = 1$.

Right-hand value: $c - 1$.

Set equal:

$$c - 1 = 1$$

$$c = 2$$

Answer: $c = 2$.

Challenge

13. $f(x) = \begin{cases} ax + b, & x < 0 \\ x^2 + 1, & 0 \leq x \leq 2, \text{ with } f(-2) = 7. \text{ Find } a \text{ and } b. \\ 3x - 1, & x > 2 \end{cases}$

Step 1 — continuity at $x = 0$:

Left-hand limit (use $ax + b$): $a(0) + b = b$.

Value at $x = 0$ (use the middle piece, since $0 \leq x \leq 2$ includes 0): $0^2 + 1 = 1$.

Set equal: $b = 1$.

Step 2 — use the extra condition $f(-2) = 7$:

Since $-2 < 0$, we use the first piece: $f(-2) = a(-2) + b = -2a + b$.

Plug in $b = 1$:

$$-2a + 1 = 7$$

$$-2a = 6$$

$$a = -3$$

Bonus check at $x = 2$: middle piece value: $2^2 + 1 = 5$. Right-hand limit (use $3x - 1$): $3(2) - 1 = 5$. They already match — this breakpoint takes care of itself no matter what a and b are, since neither piece there involves a or b .

Answer: $a = -3$, $b = 1$.

$$14. f(x) = \begin{cases} \frac{x^2-9}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}, \text{ find } k \text{ for continuity at } x = 3.$$

The limit (approaching from either side, using the “ $x \neq 3$ ” piece) requires factoring first, since plugging in $x = 3$ gives $\frac{0}{0}$:

$$\frac{x^2-9}{x-3} = \frac{(x-3)(x+3)}{x-3} = x+3$$

$$\lim_{x \rightarrow 3} (x+3) = 6$$

For continuity, we need $f(3) = k$ to match this limit:

$$k = 6$$

Answer: $k = 6$. This patches the hole that was sitting at $x = 3$.

$$15. f(x) = \begin{cases} \sqrt{x+4}, & x \leq 0 \\ 2x+a, & x > 0 \end{cases}, \text{ find } a \text{ for continuity at } x = 0.$$

Left-hand value (use $\sqrt{x+4}$, since $x \leq 0$ includes 0): $\sqrt{0+4} = \sqrt{4} = 2$.

Right-hand limit (use $2x+a$): $2(0) + a = a$.

Set equal:

$$a = 2$$

Answer: $a = 2$.

$$16. f(x) = \begin{cases} x+1, & x < 2 \\ 5, & x = 2, \text{ continuity at } x = 2? \\ x+3, & x > 2 \end{cases}$$

$f(2) = 5$, defined.

Left-hand limit: $2+1 = 3$.

Right-hand limit: $2+3 = 5$.

Since $3 \neq 5$, the two one-sided limits disagree, so the overall limit does not exist.

Answer: f is discontinuous at $x = 2$. This is a jump discontinuity — even though $f(2) = 5$ happens to match the right-hand side, the left and right limits themselves don’t agree with each other, so there’s no way to patch this by simply redefining the single point $f(2)$. The two sides are genuinely heading toward different heights.

$$17. f(x) = \frac{1}{(x-4)^2}, \text{ continuity at } x = 4?$$

$f(4)$ would require dividing by 0, so it’s undefined — condition 1 already fails.

Answer: f is discontinuous at $x = 4$. Since $(x - 4)^2$ is always positive and shrinks toward 0 from both sides as $x \rightarrow 4$, the fraction $\frac{1}{(x-4)^2}$ grows without bound toward $+\infty$ on both sides. This is an infinite discontinuity (a vertical asymptote at $x = 4$).

Applied

$$18. T(x) = \begin{cases} 0.10x, & x \leq 50 \\ 0.12x - k, & x > 50 \end{cases}, \text{ find } k \text{ for continuity at } x = 50.$$

Left-hand value: $0.10(50) = 5$.

Right-hand limit: $0.12(50) - k = 6 - k$.

Set equal:

$$6 - k = 5$$

$$k = 1$$

Answer: $k = 1$. With this value, the tax owed transitions smoothly across the $x = 50$ bracket boundary with no sudden jump.

$$19. C(w) = \begin{cases} 5 + 2w, & w \leq 3 \\ b + 3w, & w > 3 \end{cases}, \text{ find } b \text{ for continuity at } w = 3.$$

Left-hand value: $5 + 2(3) = 5 + 6 = 11$.

Right-hand limit: $b + 3(3) = b + 9$.

Set equal:

$$b + 9 = 11$$

$$b = 2$$

Answer: $b = 2$. With this value, the shipping price transitions smoothly at $w = 3$ pounds.

$$20. D(x) = \begin{cases} 20, & x \leq 1 \\ 20 + m(x - 1), & x > 1 \end{cases}, \text{ with } D(5) = 52, \text{ find } m.$$

Plug $x = 5$ into the second piece (since $5 > 1$):

$$20 + m(5 - 1) = 52$$

$$20 + 4m = 52$$

$$4m = 32$$

$$m = 8$$

Answer: $m = 8$.

Bonus explanation: continuity at $x = 1$ holds automatically for any value of m , because plugging $x = 1$ into the right-hand piece gives $20 + m(1 - 1) = 20 + m(0) = 20$ — the m term always vanishes at exactly $x = 1$, so the right side always matches the left side's value of 20 no matter what m turns out to be. That's why we needed a separate piece of information (the cost at 5 GB) to pin down m .

$$21. \ T(t) = \begin{cases} 2t + 50, & t \leq 4 \\ -3t + c, & t > 4 \end{cases}, \text{ find } c \text{ for continuity at } t = 4.$$

Left-hand value: $2(4) + 50 = 8 + 50 = 58$.

Right-hand limit: $-3(4) + c = -12 + c$.

Set equal:

$$-12 + c = 58$$

$$c = 70$$

Answer: $c = 70$. With this value, the temperature model transitions smoothly at $t = 4$ hours with no sudden jump.